

ORACLE

Cloud Operations for ExaDB-D

Change Operations Scenarios

Operations Advisory Team

Office of CTO | EMEA Solution Engineering

July, 2026

Agenda

- Introduction
- Change Operations Scenarios
 - Change Compartment
 - Scale Exadata Infrastructure
 - Scale VM Cluster
 - Resource State Management
 - Data Guard Operations



Introduction

Introduction



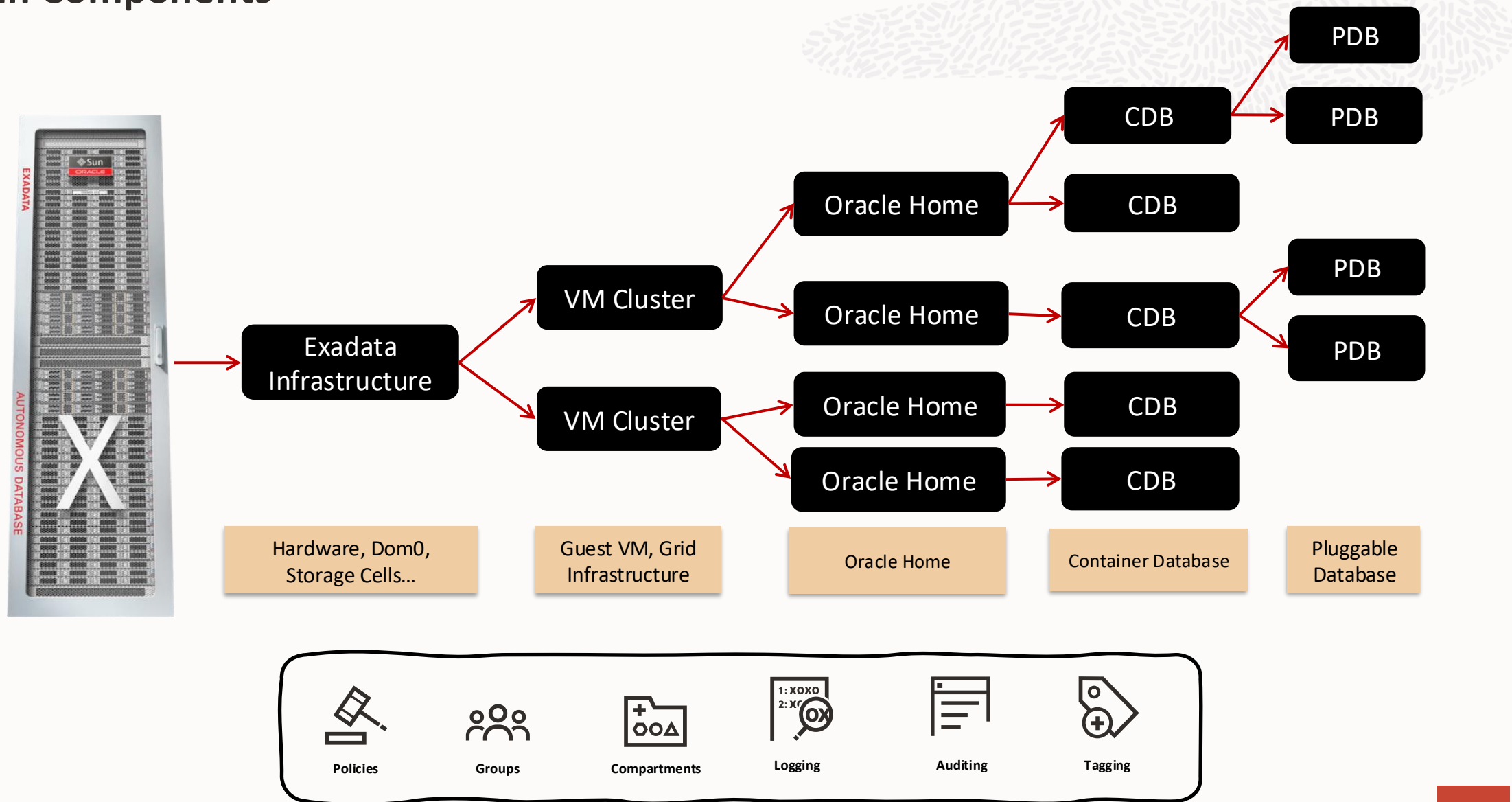
This Workshop provides information about Exadata Database Service on Dedicated Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VMC, OH, DBs, DG, scaling and patching

The code in this Workshop is just sample code and can be incomplete.

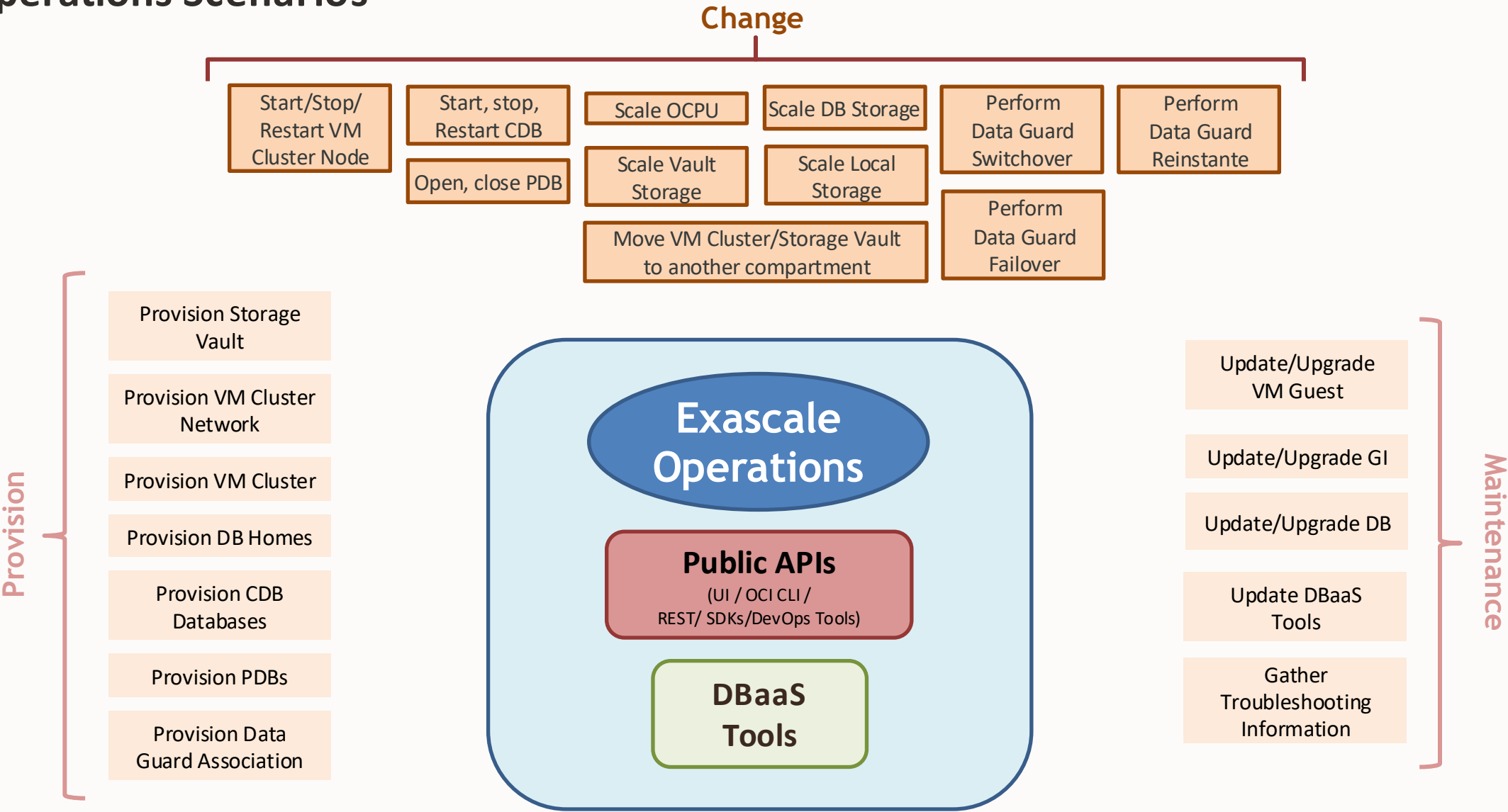
To make the sample code work in your Oracle Cloud tenancy, please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

Main Components



Change Operations

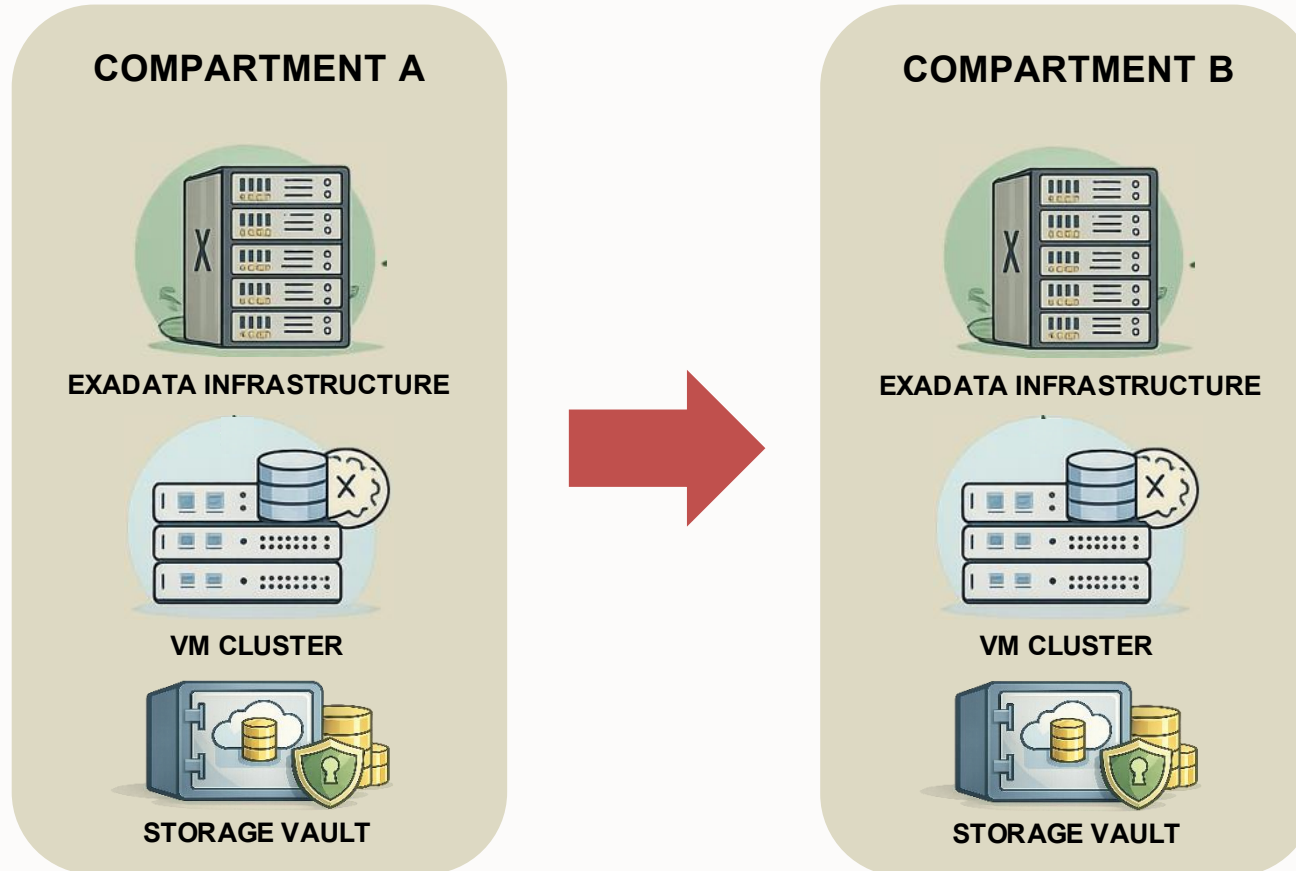
Operations Scenarios



Change compartment

Move ExaDB-D Resources Across Compartments

Flexible Shape Systems (X8M, X9M and X11M)



Key Considerations

- Resource OCID remains unchanged
- The resource enters the **Updating** state during the move.
- Destination compartment IAM policies take effect immediately.
- User access may change based on the destination policies.
- Moving a resource does not modify the Exadata topology or configuration.

How can you move resources?

You can move resources by using:

- OCI Console, OCI-CLI, REST API, SDK, IaC*

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure →
[Exadata Infrastructure | Exadata VM Cluster | Exascale Storage Vaults (inside an Exadata Infrastructure)] →
Select Actions (3 dots) on the resource → **Move resource**

Command Line Interface (OCI CLI)

Move an Exadata Infrastructure

```
oci db cloud-exa-infra change-compartment --compartment-id  
<compId> --cloud-exa-infra-id <Infra-Id> [OPTIONS]
```

Move a VM Cluster

```
oci db cloud-vm-cluster change-compartment --compartment-id  
<compId> --cloud-vm-cluster-id <VMC-Id> [OPTIONS]
```

Move a Storage Vault

```
oci db exascale-db-storage-vault change-compartment --  
compartment-id <compId> --exascale-db-storage-vault-id  
<Storage-Vault-Id> [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

compartment_id is updatable for:

- oci_database_cloud_exadata_infrastructure
- oci_database_cloud_vm_cluster
- oci_database_exascale_db_storage_vault

Scale Exadata Infrastructure

Scale Exadata Infrastructure

Flexible Shape Systems (X8M, X9M and X11M)



Infrastructure scaling expands the resource pool but does not automatically increase VM Cluster capacity.



DATABASE SERVERS

Changes:

- Compute capacity
- VM capacity



Characteristics:

- Modified online – no database downtime
- Additional capacity must be allocated to VM Cluster to be used
- Will be removed if there are no VMs running on them.

Limits:

- From 2 to 32 Database Servers



STORAGE SERVERS

Changes:

- Storage capacity
- Storage throughput
- I/O performance



Characteristics:

- Modified online – no database downtime
- Additional capacity must be assigned to VM Clusters before use.
- Remaining space should be sufficient to hold the data and ASM rebalancing.

Limits:

- From 3 to 64 Storage Servers

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure →
Exadata Infrastructure → **Scale infrastructure**

Command Line Interface (OCI CLI)

Scale Exadata Infrastructure

```
oci db cloud-exa-infra update \  
  --cloud-exa-infra-id <Infra-Id> \  
  --compute-count [integer] \  
  --storage-count [integer] [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_cloud_exadata_infrastructure"  
  "test_cloud_exadata_infrastructure" {  
    compartment_id = var.compartment_id  
    display_name =  
var.cloud_exadata_infrastructure_display_name  
    ...  
    compute_count =  
var.cloud_exadata_infrastructure_compute_count  
    storage_count =  
var.cloud_exadata_infrastructure_storage_count  
    ...  
  }
```

Scale Storage Vault - OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure →
Select the Exadata Infrastructure → Exascale Storage Vaults → Select Storage Vault → **Scale storage vault**

Command Line Interface (OCI CLI)

Scale Exascale Storage Vault

```
oci db exascale-db-storage-vault update \  
  --exascale-db-storage-vault-id <ExascaleVault_Id> \  
  --high-capacity-storage-gb [integer] [OPTIONS]
```

Terraform - Oracle Cloud Infrastructure Provider

```
resource "oci_database_exascale_db_storage_vault"  
  "test_exascale_db_storage_vault" {  
    compartment_id = var.compartment_id  
    display_name =  
var.exascale_db_storage_vault_display_name  
    ...  
    high_capacity_database_storage {  
      total_size_in_gbs =  
var.exascale_db_storage_vault_high_capacity_database_stor  
age_total_size_in_gbs  
    }  
    ...  
  }
```

Scale VM Cluster

Scale Exadata VM Cluster

Flexible Shape Systems (X8M, X9M and X11M)



COMPUTE RESOURCES

CPU

- Scaled symmetrically across all VM Cluster nodes.
- Online operation. No downtime.
- Total CPU allocation cannot exceed available Infrastructure capacity.
- To STOP all the nodes in the cluster set the OCPU to 0.

MEMORY

- Scaled symmetrically across all VM Cluster nodes.
- Compute nodes are rebooted in a rolling fashion during memory resize.
- Minimum memory per VM is 30 GB.
- Maximum memory limited by the database server resources.

LOCAL FILE SYSTEM

- Compute nodes are rebooted in a rolling fashion during local file system resize.
- Not all the file system can be changed. Check documentation.



STORAGE RESOURCES

ASM

- Online operation.
- ASM rebalance occurs automatically.
- Performed from the VM Cluster scaling workflow.
- Requires available storage capacity in the underlying Exadata Infrastructure.
- DATA and RECO allocations are managed together during the scaling workflow.

STORAGE VAULT

- Online operation.
- Elastic growth model.
- No ASM rebalance.
- Requires available capacity in the Exadata Infrastructure.
- Vault capacity and VM Cluster consumption are separate concepts.

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure →
Exadata VM Cluster → Select the VM Cluster → **Scale VM Cluster**

Command Line Interface (OCI CLI)

Scale Exadata Infrastructure

```
oci db cloud-vm-cluster update \  
  --cloud-vm-cluster-id <VMC-Id> \  
  --compute-nodes [complex type] \  
  --cpu-core-count [integer] \  
  --data-storage-size-in-tbs [text] \  
  --db-node-storage-size-in-gbs [integer] \  
  --file-system-configuration-details [complex type] \  
  --memory-size-in-gbs [integer] \  
  --ocpu-count [float] [OPTIONS]
```

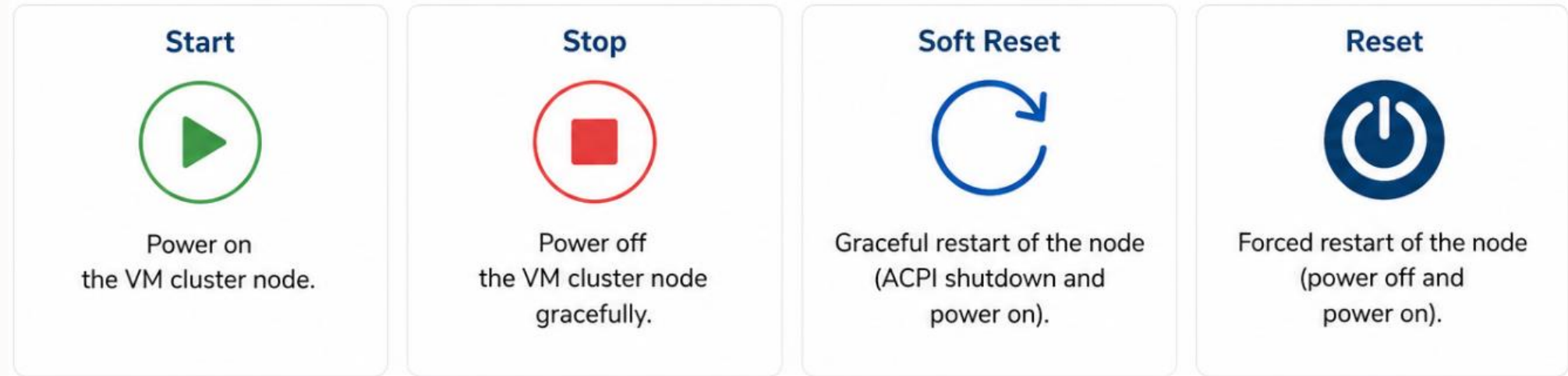
Terraform - Oracle Cloud Infrastructure Provider

For the resource `oci_database_cloud_vm_cluster` we can update the following attributes among others:

- `data_storage_percentage`
- `db_node_storage_size_in_gbs`
- `db_servers`
- `file_system_configuration_details`
 - `file_system_size_gb`
 - `mount_point`
- `memory_size_in_gbs`
- `ocpu_count`
- ...

Resource State Management

Start/Stop/Reboot VM Cluster Node



OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Virtual Machines → Select Actions (3 dots) on the VM → **Start/Stop**

OCI CLI

```
oci db node start/stop/reset/soft-reset --db-node-id <node-id> [OPTIONS]
```

Database State Operations

Container Database (CDB)

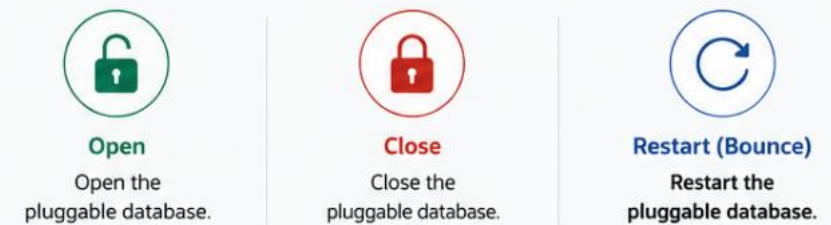


dbaascli

```
$ dbaascli database [open | close | bounce]
               [--dbname <value>] [OPTIONS]

$ dbaascli database status [--dbname <value>] [OPTIONS]
$ dbaascli database getDetails --dbname <value>
```

Pluggable Database (PDB)



OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure → VM Clusters → Container databases → Select CDB → Pluggable Databases → Select Actions (3 dots) on the PDB → Start/Stop

OCI CLI

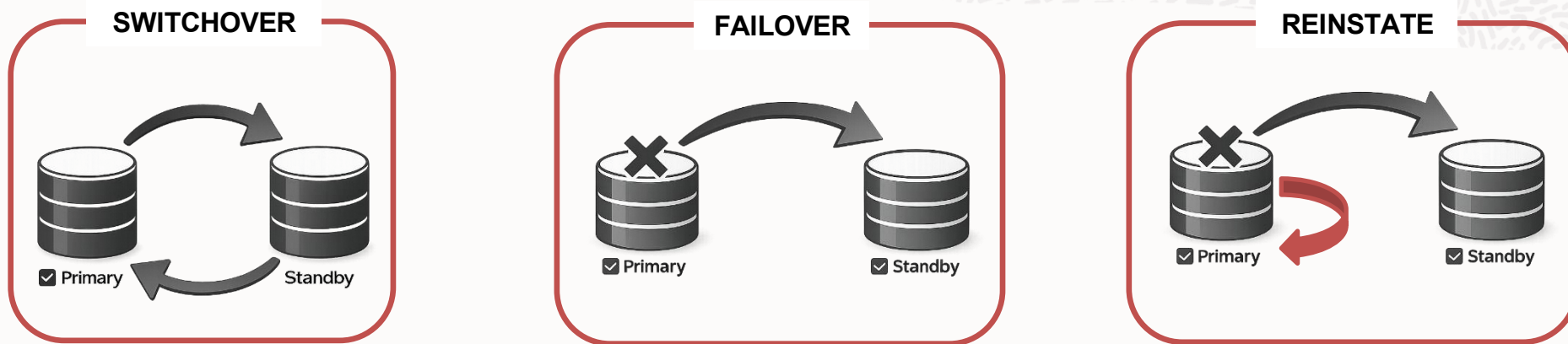
```
oci db pluggable-database [start | stop]
    --pluggable-database-id <pdb-id> [OPTIONS]
```

dbaascli

```
$ dbaascli pdb [open | close | bounce] [OPTIONS]
$ dbaascli pdb getDetails --dbname <value> [OPTIONS]
```

Data Guard Operations

Data Guard Operations



OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure → VM Clusters → Container databases → Select CDB → Data Guard Associations → Select Actions (3 dots) on the Member → **Switchover/Failover/Reinstate**

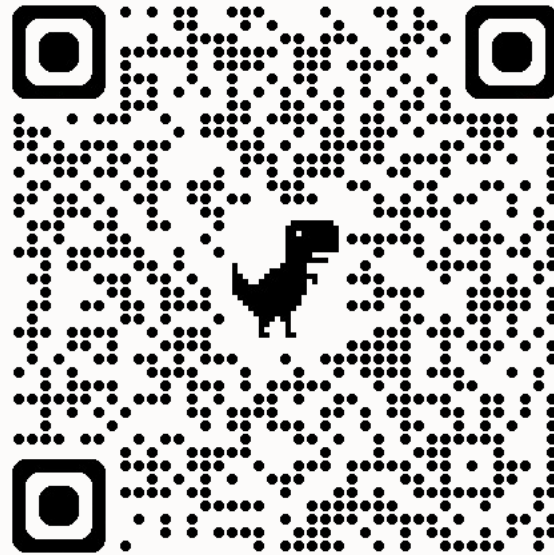
OCI CLI

```
oci db data-guard-association [switchover | failover | reinstate] --data-guard-association-id <data-guard-association-id> [OPTIONS]
```

dbaascli

```
$ dbaascli dataguard [switchover | failover | reinstate] --dbname <value> [OPTIONS]
```

Stay tuned



Operations Advisory
GitHub Repository



Thank you

ORACLE